

Comparative Evaluation of Machine Learning Models for Traffic Flow Prediction in Smart Cities

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Abstract. Bogotá has ranked among the worst-performing Latin American capitals in urban mobility for years, yet only recently has its northern corridor gained real-time traffic instrumentation. We report here on an empirical study that uses the Siemens C900 detector data collected by the Secretaría Distrital de Movilidad (SDM) between July and October 2025 — 262,271 with records across 39 access which points at 13 signalized intersections along Carreras 7 and 9 between Calle 134 and Calle 170. We trained and tested five models - Ridge Regression, Random Forest, XGBoost, a feedforward ANN, and an LSTM - on 21 features built from autoregressive lags, rolling statistics, and cyclical temporal encodings. To avoid reporting a lucky result, each model was re-run 30 times with different random seeds. XGBoost model achieved the better results metrics ($R^2 = 0.968$, RMSE = 350 veh/h), being Random Forest the second better one with ($R^2 = 0.955$). The LSTM, which most practitioners would expect to excel on a sequential task, finished last with ($R^2 = 0.850$) and showed run-to-run variance nearly 40 times larger than the tree-based models. The instability comes from a problem with the amount of data: the LSTM can't reliably estimate its gate parameters with only 744 hourly observations per sensor. In the meantime, two features alone—detection counts at t-24 h and t-168 h—made up more than 67% of XGBoost's predictive gain. This shows that daily and weekly commuting cycles are the main drivers of flow in this corridor. For transport authorities using newly installed sensor networks, these results make a clear recommendation: when historical data is short, gradient-boosted tabular models are more accurate and much more reliable than recurrent architectures.

Keywords: Traffic flow forecasting · Machine learning · XGBoost · LSTM · Smart mobility · Bogotá.

1 Introduction

For the developing world, urban mobility remains one of the most critical challenges in metropolitan areas. According to the IESE Cities in Motion Index, Bogotá ranked 174th in mobility and transport in 2025, continuing a sustained downward trend from 117th in 2019 [10]. Improving traffic management in such contexts is an important core goal of the Smart City paradigm [7,14], which

leverages Information and Communication Technologies (ICT) to take action based on data acquired from urban sensing infrastructure devices.

Short-term traffic flow forecasting -predicting vehicle counts from minutes to an hour ahead- is a fundamental building block of today’s Intelligent Transportation Systems [2,16], feeding adaptive signal-control algorithms, traveler information systems, and strategic planning tools. Over the past decade, machine learning (ML) and deep learning (DL) methods have progressively displaced classical statistical approaches [21,8,20], often showing performance gains on benchmarks such as PeMS-04, PeMS-08, and METR-LA [12,11]. Most of this literature, however, is grounded in datasets covering months or years of continuous observations across hundreds of sensors, when available over the streets or roads.

Cities in emerging economies typically operate under a very different regime: sensor coverage is recent or limited, historical records are short or not accurate, sensor-quality issues are common. Under these conditions, the relative ranking of ML and DL methods observed on large benchmarks may not hold, and a traffic-management authority with only a few weeks of data needs to choose a model without the benefit of extensive tuning.

This paper addresses this question through an empirical study in the northern corridor of Bogotá, Colombia. The Bogotá District Secretariat of Mobility (Secretaría Distrital de Movilidad, SDM) operates a network of Siemens C900 detectors at signalized intersections along the Carrera 7 and Carrera 9 arterials from calle 134 and calle 170. We analyze one month (July 2025) of real hourly data from 13 intersections (39 access points) to compare five ML models representative of the spectrum from linear baselines to deep sequential architectures: (i) Ridge Regression, (ii) Random Forest, (iii) XGBoost, (iv) a feedforward neural network (ANN), and (v) a Long Short-Term Memory (LSTM) network. Each model is trained and evaluated over 30 independent runs for robustness, and performance is reported using MAE, RMSE, MAPE, and R^2 . This analysis makes four contributions to this issue. First, we compare five popular ML models on real SDM sensor data from Bogotá. Second, the results show that gradient-boosted tabular models are much better than deep recurrent architectures in terms of both accuracy and run-to-run stability when there isn’t a lot of data. Third, the analysis of feature importance shows that two autoregressive lags—one for daily cycles (t-24 h) and one for weekly cycles (t-168 h)—together make up more than two-thirds of the model’s predictive power. Cyclical calendar features, on the other hand, only add a small amount. Fourth, a per-intersection error analysis measures how well each model works across different access points. This gives useful information for using the models in short-term operational situations.

2 Related Work

Short-term traffic flow forecasting is a crucial aspect for Smart Cities and has a rich history encompassing statistical, machine learning, and deep learning methodologies. A comprehensive review of deep learning and classical traffic forecasting models [2] emphasizes that model selection depends on temporal granularity, forecast horizon, and network characteristics. Autoregressive models (ARIMA, SARIMA) and Kalman filters [6] are used in early approaches.

These work well with smooth stationary series, but have trouble with the nonlinearities that come up in congestion regimes and scenarios. Tree ensembles became popular after that because they could handle nonlinearities naturally. Gradient-boosted tree ensembles have been shown to consistently outperform linear baselines for traffic prediction [8], and Random Forest has been successfully applied to short-term forecasting due to its robustness and interpretability [13].

The rise of deep learning brought recurrent and convolutional architectures to traffic flow forecasting. Deep learning models have demonstrated superior performance over classical methods for large-scale traffic prediction [21]. LSTM networks have been widely adopted for their ability to capture long-range temporal dependencies [20]. A comparative study on real-world traffic corridors [3] found that feature-rich inputs are more important than architectural sophistication. More recent work explores spatio-temporal graph neural networks [12,11] that jointly model spatial and temporal correlations, achieving state-of-the-art performance on PeMS and METR-LA but typically requiring at least two months of five-minute observations per sensor.

Compared with the extensive literature on Asian, European, and North American cities, studies focused on Latin American corridors remain scarce. Recent Colombian master’s theses have explored ML applications in agriculture and urban problems [17,18,19], but a comparative study on signalized-intersection traffic flow in Bogotá using real SDM data has not, to our knowledge, been published. Our work differs from prior literature in three ways: it targets an underrepresented Latin American corridor, it deliberately operates under a limited-data regime representative of recently deployed sensor networks, and it evaluates both classical and deep models within a single consistent protocol.

3 Data

3.1 Study Area and Data Source

The study area comprises 13 signalized intersections in the northern corridor of Bogotá, distributed along Carrera 7 and Carrera 9 between Calle 134 and Calle 170. These intersections carry substantial commuter traffic and serve as a major connector toward exit routes out of the city. Raw data was provided by the Bogotá District Secretariat of Mobility (SDM). Each intersection is equipped with a Siemens C900 sensor located on top of the traffic light at the approach lanes of each access. Each device reports aggregate counts and occupancy percentages at sub-hourly intervals. The dataset covers July-October, 2025 and contains 262,271 raw records across 39 unique (intersection, access) pairs.

3.2 Preprocessing and Aggregation

Raw records have timestamps that align by hour, but have 1–4 sub-hourly reports per (hour, EXT, Access) combination. We further aggregate these reports into single value per hour per access based on these rules: detections are the sum of the reports for the hour, and occupancy the mean of the reports, weighted by the detection counts. The result is a clean time series of hourly data with 744 observations per access (31 days $\times$ 24 hours) for a total of 29,016 rows. The sporadic sensor downtime (around 8% of all slots) causes missing data,

but this data is imputed using linear temporal interpolation within the series. Occupancy values are also clipped to the physical limits of $[0, 100]\%$. We also adjust detection values in order to remain consistent with a flow of zero. There are some remaining metadata fields (cardinal, turn type, street level) that are appended from SDM using a lookup table.

3.3 Period Selection

The initial exploration of the SDM extract from July–October 2025 indicates a considerable structural decrease in registered flow starting from August 1, 2025, with a subsequent monotonic decrease of approximately 70% by the end of October. The suddenness of this change – a discrete step at the month boundary with no gradual drift – is incompatible with an intrinsic change in traffic behavior, and consistent with a systemwide recalibration of the sensors, or a modification of the SDM reporting. In order to prevent the training from being contaminated by this nonstationarity, we limit our analyses to July 2025, the last month with consistent reporting.

3.4 Descriptive Analysis

Figure 1 shows the mean hourly detection across the corridor as a function of hour-of-day and day-of-week. Three patterns are immediately visible: first, weekdays exhibit the expected bimodal structure with peak traffic concentrated between 16:00 and 18:00. Second, weekends display a different signature dominated by a midday plateau, reflecting commercial and leisure activity in the corridor’s surroundings. Third, night hours (00:00–05:00) show uniformly low activity regardless of the day.

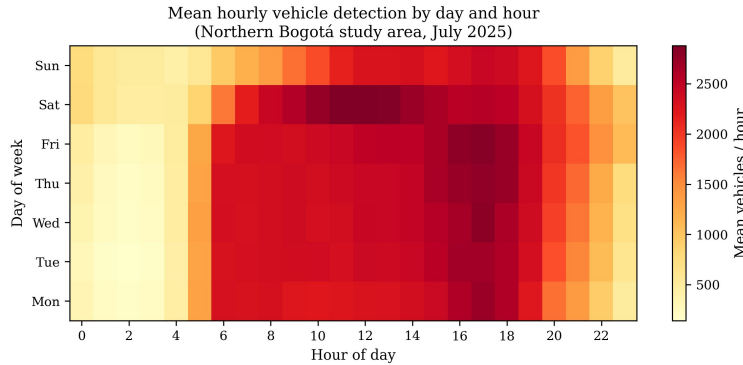


Fig. 1. Mean hourly vehicle detection by day of week and hour of day across the study corridor. Darker cells indicate higher mean flow.

Figure 2 contrasts the daily profile on weekdays versus weekends. On weekdays, flow rises sharply at 06:00 and sustains a plateau until a pronounced evening peak near 17:00, consistent with commuter behavior. On weekends, the rise is more gradual and the afternoon peak is both lower in magnitude and earlier in time. Notably, on weekend midday the corridor briefly exceeds weekday midday flow, likely reflecting shopping and recreational traffic along Carrera 7.

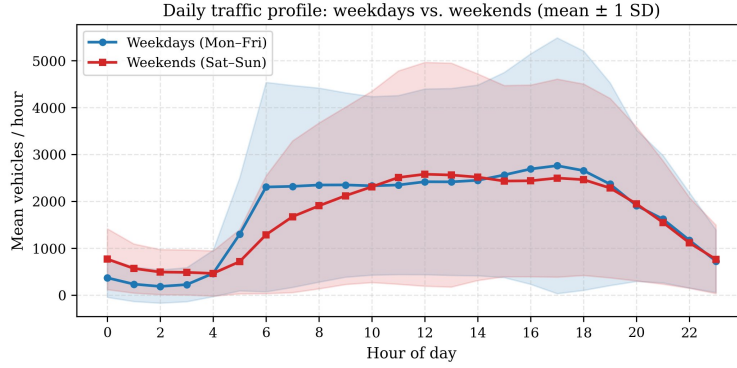


Fig. 2. Daily traffic profile: weekdays (blue) versus weekends (red). Shaded bands indicate ± 1 standard deviation across all 39 access points.

Figure 3 presents three representative time series illustrating the heterogeneity of the corridor: a high-flow access (EXT 1178, Access 2), a medium-flow access (EXT 1016, Access 4), and a degenerate low-flow series (EXT 1169, Access 4) corresponding to a sensor with a nearly flat reading profile. The inclusion of the last series illustrates sensor-quality challenges that are typical of operational SDM data and are discussed in Section 6.

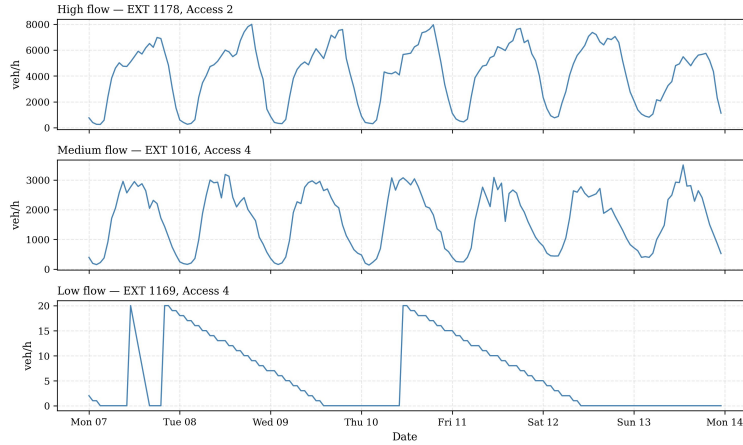


Fig. 3. Example time series for three access points of contrasting flow levels (one representative week).

Overall, the aggregated dataset spans $1,196 \pm 1,472$ vehicles per hour (mean $\pm$ SD), with a median of 622 veh/h and a 99th percentile of 6,359 veh/h, reflecting the high heterogeneity of flow regimes across the corridor. Detection and occupancy exhibit a moderate nonlinear relationship (Pearson $r = 0.47$, Spearman $\rho = 0.62$).

4 Methodology

4.1 Problem Formulation

Let $y_{s,t}$ denote the hourly vehicle detection count at access point $s \in \{1, \dots, 39\}$ and hour t . The forecasting task is to predict $y_{s,t+1}$ given a vector of features $\mathbf{x}_{s,t}$ derived from observations up to and including hour t :

$$\hat{y}_{s,t+1} = f_{\theta}(\mathbf{x}_{s,t}), \quad (1)$$

where f_{θ} denotes a machine-learning model with parameters θ estimated from the training data. The same model is trained jointly over all 39 series rather than fitting one model per access, enabling cross-series learning and reducing the risk of overfitting on short series.

4.2 Feature Engineering

For each (series, time) pair we construct 21 features grouped into four families. *Autoregressive lags of detection* (7 features): `det_lag_1`, `det_lag_2`, `det_lag_3`, `det_lag_6`, `det_lag_12`, `det_lag_24`, and `det_lag_168`, covering recent, intraday, and weekly cycles. *Rolling aggregates* (3 features): 3-hour and 24-hour rolling means of detection (`det_ma_3`, `det_ma_24`) and 24-hour rolling standard deviation (`det_std_24`), all computed over strictly past values to avoid target leakage. *Cyclical temporal encodings* (7 features): sine and cosine of hour-of-day and day-of-week, a binary weekend indicator, and two peak-hour flags (`is_peak_morning` for 06:00–09:00 weekdays and `is_peak_evening` for 16:00–19:00 weekdays). *Contemporaneous and spatial features* (4 features): one-hour lagged occupancy (`ocu_lag_1`), current occupancy at t , and the integer identifiers of intersection (EXT) and access point (ACCESO) as categorical codes. The first 168 hours of each series are discarded as warm-up because `det_lag_168` is undefined during that period, resulting in 22,425 usable observations.

4.3 Temporal Split

We apply a strictly chronological train/validation/test split at the timestamp level: *Training* (July 8–20, 12,168 observations, 54%); *Validation* (July 21–25, 4,680 observations, 21%); *Testing* (July 26–31, 5,577 observations, 25%). All model selection, hyperparameter tuning, and early-stopping decisions use the validation set; the test set is touched only for final reporting.

4.4 Models

We compare five models spanning the spectrum from linear to deep sequential. **Ridge Regression** (baseline) uses L^2 regularization with α selected via 5-fold time-series cross-validation over $\{0.01, 0.1, 1, 10, 100\}$; features are standardized. **Random Forest** [4] uses 300 trees, `max_depth=20`, and `min_samples_leaf=2`, without feature scaling. **XGBoost** [5] uses up to 500 rounds, learning rate 0.05, `max_depth=8`, subsample and column-sample rates of 0.8, L^1/L^2 regularization ($\alpha = 0.1$, $\lambda = 1.0$), and early stopping after 20 rounds without validation improvement. **ANN** is a three-layer MLP of sizes 128/64/32 with ReLU, Batch-Norm, and dropout rates 0.2/0.2/0.1, trained with Adam (10^{-3}), batch size 512,

up to 150 epochs, early stopping (patience 15), and learning-rate reduction on plateau (patience 7, factor 0.5). The target is log-transformed via $\log(1 + y)$ and inverted at inference. **LSTM** [9] is a two-layer stacked recurrent network with 64 and 32 hidden units, a 16-unit dense layer, and a linear output. The input is a sliding window of 24 hours of the 21-feature vector; the optimizer, callbacks, and log-transform match the ANN, with batch size 256 and up to 100 epochs.

4.5 Evaluation Protocol

For each model we perform 30 independent training runs with different random seeds to capture run-to-run variability. For Ridge, variability is induced by bootstrap resampling of the training set. For Random Forest, XGBoost, ANN, and LSTM, it is induced by the respective random-state parameters of the underlying libraries.

Each run reports four metrics computed on the test set:

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|, \quad \text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}, \quad (2)$$

$$\text{MAPE} = \frac{100}{|\mathcal{I}|} \sum_{i \in \mathcal{I}} \left| \frac{y_i - \hat{y}_i}{y_i} \right|, \quad R^2 = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y})^2}, \quad (3)$$

where $\mathcal{I} = \{i : y_i > 0\}$ is the set of observations with strictly positive ground-truth detection. MAPE is computed only on this subset because of the well-known instability of the metric when the denominator approaches zero.

All experiments were implemented in Python 3.11 using `scikit-learn` 1.4 [15], `xgboost` 2.0, and `tensorflow` 2.15 [1]. Training was performed on a workstation equipped with an Intel Core i7-12700 processor, 32 GB of DDR5 memory, and an NVIDIA RTX 4060 Ti GPU used for the ANN and LSTM.

5 Results

5.1 Overall Performance

Table 1 reports the mean and standard deviation of each metric on the test set, computed over 30 independent runs. XGBoost achieves the best performance across all four metrics, with a mean $R^2 = 0.968 \pm 0.001$ and a mean RMSE of 350 ± 5 veh/h. Random Forest is a close second ($R^2 = 0.955$), followed by the ANN ($R^2 = 0.925$). Ridge, despite its simplicity, delivers a respectable $R^2 = 0.915$, within two percentage points of the ANN. LSTM is the worst performer, with mean $R^2 = 0.850$ and a standard deviation of 0.040 —nearly two orders of magnitude larger than that of the tree-based models.

Table 1. Test-set performance comparison across the five models. Values are mean $\pm$ standard deviation over 30 independent runs. Best value in each column is shown in bold.

Model	MAE	RMSE	MAPE (%)	R^2
Ridge	344.7 $\pm$ 2.4	571.4 $\pm$ 2.1	220.4 $\pm$ 12.2	0.9148 $\pm$ 0.0006
Random Forest	201.5 $\pm$ 0.3	417.3 $\pm$ 1.2	23.1 $\pm$ 0.2	0.9546 $\pm$ 0.0003
XGBoost	181.5 $\pm$ 1.6	350.4 $\pm$ 5.3	26.8 $\pm$ 1.2	0.9680 $\pm$ 0.0010
ANN	283.7 $\pm$ 19.8	536.6 $\pm$ 32.1	26.0 $\pm$ 0.9	0.9246 $\pm$ 0.0091
LSTM	387.3 $\pm$ 59.8	752.3 $\pm$ 98.4	33.6 $\pm$ 3.8	0.8499 $\pm$ 0.0398

The abnormally large MAPE of Ridge (220%) is an artifact of linear predictions occasionally falling below zero during low-flow nighttime hours, which inflates the denominator of the relative error. The tree-based and neural models do not suffer from this problem because their predictions are naturally bounded near zero.

Figure 4 shows the predicted-versus-actual scatter for the best run of each model on the test set, using a common axis scale so that the five panels are directly comparable. XGBoost (top right) produces the tightest cloud around the $y = x$ diagonal, followed by Random Forest. Ridge (top left) displays the widest dispersion in the low-flow region and visible under predictions in the high-flow tail. The ANN panel shows slightly more scatter compared to Random Forest. LSTM shows dispersion across the entire range the widest, especially for target values above 4,000 veh/h. The visual impression is consistent with the numerical ranking in Table 1.

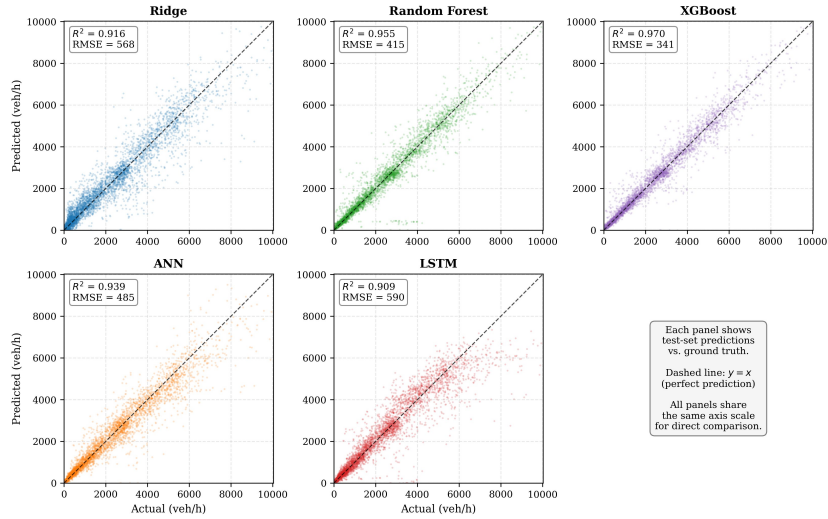


Fig. 4. Predicted vs. actual vehicle detection on the test set for the five models. Each panel shows the best run of the corresponding model; the R^2 and RMSE reported inside each panel are for that single run and may differ slightly from the mean values over 30 runs reported in Table 1.

Figure 5 give an overview op MAE, RMSE, and R squared with error bars that represent the standard deviation. The Ridge, Random Forest, and XGBoost error bars are small compared to ANN and especially LSTM error bars. This is a significant finding: deep models are less accurate and far more unreliable.

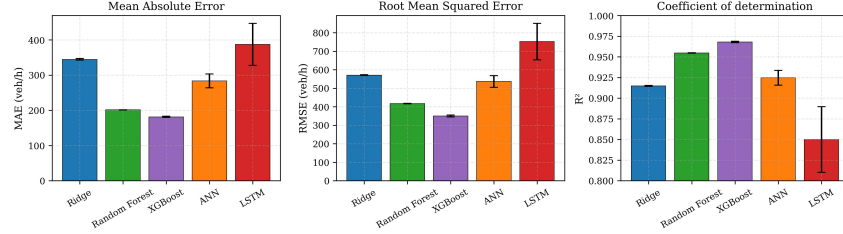


Fig. 5. Comparison of test-set metrics across the five models. Error bars show ± 1 standard deviation over 30 independent runs.

5.2 Feature Importance and the Dominance of Two Lags

Figure 6 shows the feature-importance scores of the XGBoost model (computed as gain-based importance). Two features dominate: `det_lag_24` accounts for approximately 40% of the total importance, and `det_lag_168` for an additional 27%. Together, these two autoregressive lags explain over 67% of the gain accumulated by the model. The immediate lag `det_lag_1` contributes an additional 8%; all remaining features—including the cyclical encodings—contribute less than 5% each.

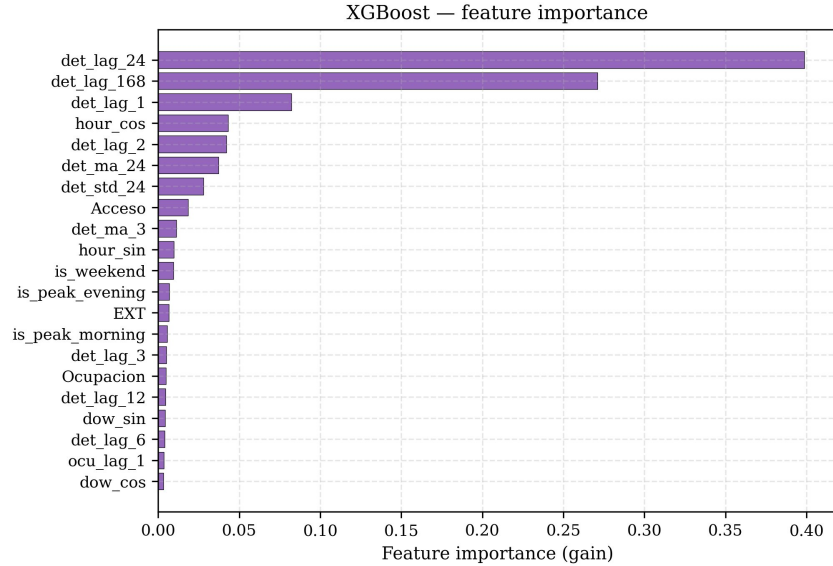


Fig. 6. Feature importance (gain) of the XGBoost model. Two lags of 24 h and 168 h jointly account for more than 67% of the total gain.

The Random Forest feature-importance profile is qualitatively similar but even more concentrated, with `det_lag_24` alone accounting for 71% and `det_lag_168` for 20%, leaving less than 9% for all other features. This concentration is consistent with the learning-curve behavior reported below and reinforces the interpretation that the predictive content of the dataset is overwhelmingly carried by the daily and weekly cycles.

5.3 Per-Intersection Analysis

Figure 7 shows the per-intersection test RMSE for each of the five models across all 13 intersections of the corridor. Three observations stand out.

First, the relative ordering of models is remarkably stable across intersections: XGBoost is at or near the top in every case except for the degenerate intersections (1169 and 1019), where all models effectively predict zero and errors are negligible by construction.

Second, intersection 1170 is an outlier: Ridge, Random Forest, ANN, and LSTM all post RMSE values in the 750–1030 veh/h range, while XGBoost achieves 630 veh/h—a 38% improvement over the next-best model. This single intersection accounts for much of the aggregate advantage of XGBoost over Random Forest.

Third, LSTM does not win on any intersection. On high-flow intersections (1178, 1229, 1837), its RMSE is 20–60% worse than XGBoost, and even on medium-flow intersections (1036, 1171) it lags the tabular models by 20–30%.

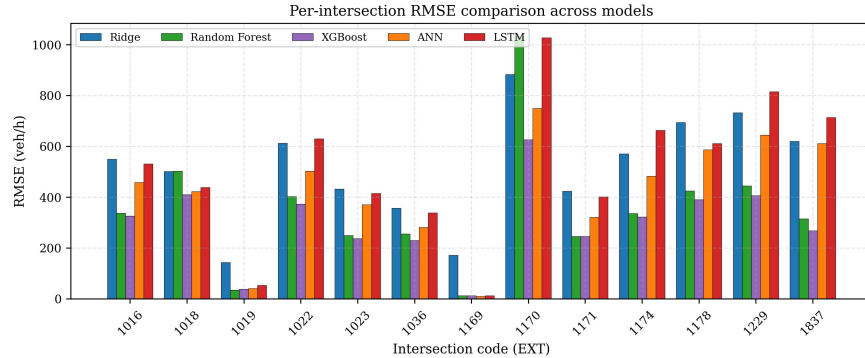


Fig. 7. Per-intersection test RMSE across all five models. Intersections are sorted by intersection code.

5.4 Convergence Behavior

The learning curves of XGBoost and LSTM (Figure 8) shed light on why the ranking observed above holds. For XGBoost, training RMSE descends monotonically throughout the 500 boosting rounds while validation RMSE stabilizes around round 100 and continues to decrease very slowly. Early stopping is triggered at round 497, indicating that XGBoost was still extracting marginal improvements at the end of the budget. For LSTM, in contrast, training loss drops sharply in the first epoch and then plateaus essentially flat for the remaining ~ 30 epochs. This behavior indicates that the LSTM is not extracting structured sig-

nal from the data beyond what can be learned in the first few passes —consistent with the hypothesis that the small number of effective training sequences per series prevents the recurrent gates from converging to a meaningful temporal representation.

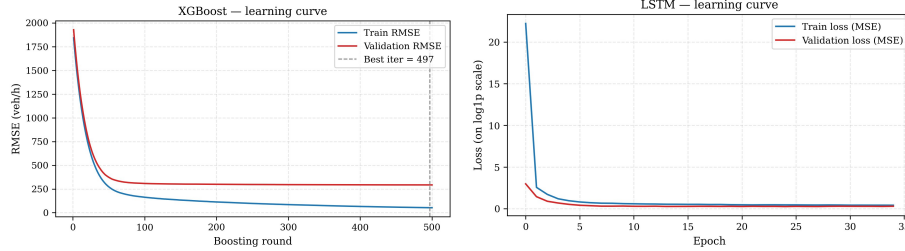


Fig. 8. Learning curves of XGBoost (left, RMSE per boosting round) and LSTM (right, MSE per epoch on the $\log(1 + y)$ scale). XGBoost continues to improve until the end of the budget; LSTM plateaus after one epoch.

6 Discussion

The finding that `det_lag_24` and `det_lag_168` jointly explain more than two-thirds of predictive importance has a natural interpretation. Urban vehicle flow is characterized by strong daily and weekly cycles driven by commuting, work schedules, and commercial activity: the 24-hour lag captures the daily cycle and the 168-hour lag the weekly one. All five models converge on this interpretation in their own way: Ridge distributes weight across them linearly, Random Forest and XGBoost encode the interaction in tree splits, and the neural models rely on them through the input vector. A consequence is that the *residual* predictive problem is small; a simple persistence model predicting $\hat{y}_{t+1} = 0.5 y_{t-23} + 0.5 y_{t-167}$ already achieves $R^2 \approx 0.89$, establishing a strong floor that any model must exceed and explaining why the margin between top performers is narrow.

The underperformance of LSTM relative to all baselines—including linear regression—is the most counterintuitive finding and deserves discussion. On standard benchmarks such as PeMS-04 and METR-LA, LSTM variants typically outperform tabular baselines, but those benchmarks share two structural features our dataset lacks: at least two months of five-minute observations ($\sim 17,000$ timesteps per sensor) and hundreds of sensors. Our dataset has only 744 hourly timesteps per series and 39 series, yielding approximately 28,000 training sequences total after 24-hour windowing. As a result the LSTM cannot estimate the stateful parameters of its gates in a stable way. The learning curve in Figure 8 shows convergence within the first epochs followed by a flat plateau—the signature of a recurrent network that has memorized coarse statistics without learning meaningful temporal dependencies. The large run-to-run variance (test RMSE ranging from 590 to 936 across 30 runs) further supports this interpretation. A practical implication follows: *in limited-data regimes, recurrent deep architectures should not be assumed to outperform tabular baselines*, and practitioners should benchmark them carefully before deploying.

Out of the five models, XGBoost gives the best trade-off between accuracy and cost. Using the hardware for this study, XGBoost completed 30 training runs in about 4 minutes. In comparison, Artificial Neural Network (ANN) model took about 15 minutes, LSTM model took about 25 minutes on the GPU, and Random Forest (RF) model took about 10 minutes on the CPU. The per-intersection analysis in Figure 7 shows model quality is not consistent in the corridor: three intersections (1169, 1019, 1170) show degenerate behavior, while the other ten have consistent model ordering. Importantly, XGBoost makes the best prediction for the challenging intersection 1170, which means that its gradient boosted tree structure is better suited for dealing with the heterogeneous noise profiles as compared to other models.

Several limitations of this study should be noted. Temporal coverage is limited to one month, which is sufficient to estimate relative model rankings under limited-data conditions but does not capture seasonal variation. Three of the 39 access points show sensor-quality issues that inflate per-intersection RMSE dispersion; we retained them to reflect real operational conditions. MAPE is reported only on strictly positive ground-truth values to avoid numerical instability, and SMAPE is left for future work. Finally, the study is restricted to a single corridor of Bogotá, and generalization to other corridors or cities remains an open question.

7 Conclusions

This paper evaluated five models to forecast traffic flow one hour ahead in a 13-intersection corridor in northern Bogotá using one month of real SDM sensor data. The best performing model was XGBoost with an $R^2 = 0.968$. In second place was Random Forest (0.955) followed by ANN (0.925), Ridge (0.915), and LSTM (0.850). LSTM performed the worst which is counterintuitive to the authors since the task is sequential, but this was attributed to the small number of training sequences available. Feature-importance indicated that lagged predictors of 24 hours and 168 hours (weekly) explain the most (over 2/3) of the variability (predictive signal). This shows that urban traffic flow has significant daily and weekly seasonality.

The authors concluded that if data history is short and sensor coverage is limited, use of tabular gradient-boosted models is the best approach, and reserve recurrent deep models to cases with much larger data sets. This study will expand in three ways; (i) temporally, once SDM sensors are fixed and more data can be acquired, (ii) exogenously, adding weather and calendars, (iii) spatio-temporally, graph neural networks when data is available.

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